COURSE 1 · MATHEMATICS FOUNDATIONS

Mathematics foundations

Printable flashcards

141 cards: equations, key ideas, quick-check questions and problems.
Cut, fold and review; or use the scheduled version in the app.

How to use

Printable flashcards

Each row is one card: the prompt on the left, the answer on the right. Print single-sided, cut along the rows, and fold each strip down the middle so the answer sits on the back. The same cards are in the app, where they are scheduled for review automatically.

Module 1.1 · Numbers and arithmetic fluency

EQUATION Write the equation: Percentage change	$\%\ \text{change} = \frac{\text{new} - \text{old}}{\text{old}} \times 100$
EQUATION Write the equation: Multiplying powers	$(a \times 10^m)(b \times 10^n) = ab \times 10^{m+n}$
EQUATION Write the equation: Dividing powers	$\frac{a \times 10^m}{b \times 10^n} = \frac{a}{b} \times 10^{m-n}$
EQUATION Write the equation: Negative and zero powers	$10^{-n} = \frac{1}{10^n}, \qquad 10^0 = 1$
KEY IDEA Complete the key idea: Percentage change is …	Percentage change is measured against the starting value
KEY IDEA Complete the key idea: In scientific notation, …	In scientific notation, multiply coefficients and add exponents
KEY IDEA Complete the key idea: Estimate first; it tells …	Estimate first; it tells you whether a calculated answer is sensible
KEY IDEA Complete the key idea: Report results to the …	Report results to the precision of your least precise input
QUICK CHECK What is 10^{-3} as a fraction?	1/1000
QUICK CHECK $(2 \times 10^3) \times (3 \times 10^4)$ equals	6×10^7
QUICK CHECK A price falls 50% then rises 50%. It ends at	75% of the original. The rise is 50% of a smaller base.

QUICK CHECK The prefix 'micro' means	10^{-6}
PROBLEM A jacket priced at $180 is reduced by 35%. What is the sale price?	117 dollars. 180 − 63 = 117. Or directly: 0.65 × 180 = 117.
PROBLEM A town's population grows from 24 000 to 27 600. What is the percentage increase?	15 %. 3600 ÷ 24 000 = 0.15, which is 15%.
PROBLEM Support for a policy rises from 40% to 50%. Which statement is correct?	It rose by 10 percentage points, or 25%. Relative to the starting 40, that is 10 ÷ 40 = 25%.
PROBLEM Calculate $(6 \times 10^{5}) \times (4 \times 10^{-2})$. Give the answer as an ordinary number.	2.4e+04. $24 \times 10^{3} = 2.4 \times 10^{4} = 24\,000$.
PROBLEM Light travels at 3.0×10^{8} m/s. The Sun is 1.5×10^{11} m away. How many seconds does sunlight take to reach Earth?	500 s. $0.5 \times 10^{3} = 500$ s, a little over eight minutes.
PROBLEM A red blood cell is about 8 micrometres across. How many would fit side by side across 1 centimetre?	1250. $10^{-2} \div (8 \times 10^{-6}) = 0.125 \times 10^{4} = 1250$.
PROBLEM A calculator gives 3.14159265 for a result whose inputs were measured to 3 significant figures. How should you report it?	3.14. 3.14.
PROBLEM Estimate, to the nearest power of ten, how many times your heart beats in an 80-year life. Show your assumptions.	About 70 beats a minute × 60 minutes × 24 hours × 365 days × 80 years. Rounding: $70 \times 60 \approx 4 \times 10^{3}$ per hour; $\times 24 \approx 10^{5}$ per day; $\times 365 \approx 3.7 \times 10^{7}$ per year; $\times 80 \approx 3 \times 10^{9}$. So roughly three billion, of order 10^{9}.

Module 1.2 · Units, measurement and dimensional analysis

EQUATION Write the equation: The newton in base units	$1\ \mathrm{N} = 1\ \mathrm{kg\,m\,s^{-2}}$
EQUATION Write the equation: Area and volume conversions	$1\ \mathrm{m^2} = 10^4\ \mathrm{cm^2}, \qquad 1\ \mathrm{m^3} = 10^6\ \mathrm{cm^3} = 1000\ \mathrm{L}$
EQUATION Write the equation: Dimensional argument for a pendulum	$[T] = \sqrt{\frac{[L]}{[g]}} = \sqrt{\frac{\mathrm{m}}{\mathrm{m\,s^{-2}}}} = \mathrm{s}$
EQUATION Write the equation: Percentage uncertainty	$\%\ \text{uncertainty} = \frac{\Delta x}{x} \times 100$
KEY IDEA Complete the key idea: Convert to SI …	Convert to SI base units before substituting
KEY IDEA Complete the key idea: Conversion factors equal one; raise …	Conversion factors equal one; raise them to the power of the unit
KEY IDEA Complete the key idea: Both sides of an …	Both sides of an equation must have the same dimensions
KEY IDEA Complete the key idea: Averaging reduces random …	Averaging reduces random error, not systematic error
QUICK CHECK The SI base unit of mass is the	kilogram
QUICK CHECK 1 m^3 equals how many cm^3?	1 000 000
QUICK CHECK Averaging many readings reduces	random error
QUICK CHECK Which quantity could be added to a speed?	another speed
PROBLEM Convert 90 km/h into metres per second.	25 m/s. 90 000 ÷ 3600 = 25 m/s.

PROBLEM A room is 4.0 m by 3.5 m. What is its floor area in square centimetres?	1.400e+05 cm^2. 1 m^2 = 10^4 cm^2, so 14 × 10^4 = 1.4 × 10^5 cm^2.
PROBLEM A tank holds 2.5 cubic metres of water. How many litres is that?	2500 L. 2.5 × 1000 = 2500 L.
PROBLEM Which of these equations is dimensionally impossible? (v speed, t time, a acceleration, s distance)	s = v + at. s = v + at adds metres to metres-per-second: impossible.
PROBLEM What are the SI base units of a joule?	kg m^2 s^{-2}. N = kg m s^{-2}, so J = kg m^2 s^{-2}.
PROBLEM A length is measured as 25.0 ± 0.5 cm. What is the percentage uncertainty?	2 %. 0.02 × 100 = 2%.
PROBLEM A friend averages 50 readings from a scale that always reads 0.2 kg too high. Explain why the average is still wrong, and what kind of error this is.	This is a systematic error: every reading is shifted by the same 0.2 kg in the same direction. Averaging cancels random scatter but not a constant offset, so the mean is still 0.2 kg too high. The fix is to calibrate the scale, for example by weighing a known mass, and correct for the offset.

Module 1.3 · Geometry

EQUATION Write the equation: Pythagoras' theorem	$a^2 + b^2 = c^2$
EQUATION Write the equation: Diagonal of a box	$d = \sqrt{a^2 + b^2 + c^2}$
EQUATION Write the equation: Scaling of length, area and volume	$L \to kL, \quad A \to k^2 A, \quad V \to k^3 V$
EQUATION Write the equation: Circle	$C = 2\pi r, \qquad A = \pi r^2$

EQUATION Write the equation: Sphere	$A = 4\pi r^2, \qquad V = \frac{4}{3}\pi r^3$
KEY IDEA Complete the key idea: Pythagoras: hypotenuse squared equals …	Pythagoras: hypotenuse squared equals the sum of the other two squares
KEY IDEA Complete the key idea: Scale lengths by k: …	Scale lengths by k: areas by k^2, volumes by k^3
KEY IDEA Complete the key idea: Circle: C = 2πr, A = …	Circle: C = 2πr, A = πr^2. Sphere: A = $4\pi r^2$, V = $(4/3)\pi r^3$
KEY IDEA Complete the key idea: Surface-to-volume ratio …	Surface-to-volume ratio falls as size grows
QUICK CHECK A right triangle has legs 6 and 8. The hypotenuse is	10
QUICK CHECK Scale lengths by 3. Volume scales by	27
QUICK CHECK The surface area of a sphere is	$4\pi r^2$
QUICK CHECK Angles in a triangle add to	180°
PROBLEM A ladder 5.0 m long leans against a wall with its foot 1.4 m from the wall. How high up the wall does it reach?	4.8 m. $h^2 = 25 - 1.96 = 23.04$, so h = 4.8 m.
PROBLEM What is the area of a circle of diameter 10 cm?	78.54 cm^2. $A = \pi \times 5^2 \approx 78.5\ cm^2$.
PROBLEM A box measures 30 cm × 40 cm × 120 cm. How long is the longest straight rod that fits inside?	130 cm. = √(900 + 1600 + 14 400) = √16 900 = 130 cm.

PROBLEM A model car is built at 1:20 scale. Its paintwork covers 0.15 m^2. What area of paint covers the real car?	60 m^2. 0.15 × 400 = 60 m^2.
PROBLEM If you double the radius of a sphere, by what factor does its volume increase?	8 ×. 2^3 = 8.
PROBLEM Earth's radius is about 6.37 × 10^6 m. Treating it as a sphere, what is its volume?	1.083e+21 m^3. r^3 = 2.58 × 10^{20}; × 4.19 gives about 1.08 × 10^{21} m^3.
PROBLEM A cube's side length triples. Its surface-area-to-volume ratio	falls to one third. Ratio scales by 9/27 = 1/3.
PROBLEM Use the square-cube law to explain why elephants have proportionally thicker legs than gazelles.	Weight goes with volume, so it scales with length cubed. The strength of a leg bone goes with its cross-sectional area, which scales with length squared. A larger animal built to the same proportions would carry far more weight per unit of bone area, so the bones would fail. To compensate, large animals need disproportionately thick legs.

Module 1.4 · Trigonometry

EQUATION Write the equation: Trigonometric ratios	$\sin\theta = \frac{\text{opp}}{\text{hyp}}, \quad \cos\theta = \frac{\text{adj}}{\text{hyp}}, \quad \tan\theta = \frac{\text{opp}}{\text{adj}}$
EQUATION Write the equation: Resolving a vector of size F at angle θ	$F_x = F\cos\theta, \qquad F_y = F\sin\theta$
EQUATION Write the equation: Degrees and radians	$\pi \text{ rad} = 180^\circ$
EQUATION Write the equation: Arc length	$s = r\theta$
EQUATION Write the equation: Pythagorean identity	$\sin^2\theta + \cos^2\theta = 1$

EQUATION Write the equation: General sinusoid	$y = A\sin(\omega t + \phi), \qquad T = \frac{2\pi}{\omega}$
KEY IDEA Complete the key idea: SOH CAH TOA in …	SOH CAH TOA in right triangles; inverse functions give angles
KEY IDEA Complete the key idea: Resolve vectors with cos for …	Resolve vectors with cos for the adjacent component and sin for the opposite
KEY IDEA Complete the key idea: π radians = 180°; arc …	π radians = 180°; arc length s = rθ (θ in radians)
KEY IDEA Complete the key idea: Sine waves: …	Sine waves: amplitude, period and phase
QUICK CHECK tan θ equals	opp/adj
QUICK CHECK 180° in radians is	π
QUICK CHECK A 10 N force at 60° to the horizontal has a horizontal component of	5 N. 10 cos 60° = 5 N.
QUICK CHECK The period of y = sin(ωt) is	2π/ω
PROBLEM A ramp rises at 12° and is 4.0 m long along its surface. How high does it rise?	0.8316 m. h = 4.0 × sin 12° ≈ 0.83 m.
PROBLEM Convert 45° to radians, as a decimal.	0.7854 rad. 45π/180 = π/4 ≈ 0.785 rad.
PROBLEM From 50 m away, the top of a tower is at an angle of elevation of 35°. How tall is the tower (ignoring eye height)?	35.01 m. height = 50 × tan 35° ≈ 35 m.

PROBLEM A rope pulls a sled with 200 N at 25° above the horizontal. What is the horizontal component of the pull?	181.3 N. 200 × cos 25° ≈ 181 N.
PROBLEM A wheel of radius 0.30 m turns through 2.5 radians. How far does a point on the rim travel?	0.75 m. 0.30 × 2.5 = 0.75 m.
PROBLEM A right triangle has opposite side 7 and hypotenuse 10. Find the angle, in degrees.	44.43 °. $\theta = \sin^{-1}(0.7) \approx 44.4°$.
PROBLEM The wave y = 3 sin(2t) has amplitude and period	3 and π. Period = $2\pi/\omega = 2\pi/2 = \pi$.
PROBLEM Explain, using the unit circle, why $\sin^2\theta + \cos^2\theta = 1$ for every angle.	On the unit circle, the point at angle θ has coordinates (cos θ, sin θ). Its distance from the centre is always the radius, 1. By Pythagoras, the horizontal and vertical distances satisfy $\cos^2\theta + \sin^2\theta = 1^2 = 1$, whatever θ is.

Module 1.5 · Probability and statistics

EQUATION Write the equation: Addition rule (mutually exclusive)	$P(A \text{ or } B) = P(A) + P(B)$
EQUATION Write the equation: Multiplication rule (independent)	$P(A \text{ and } B) = P(A)\,P(B)$
EQUATION Write the equation: Conditional probability	$P(A \mid B) = \frac{P(A \text{ and } B)}{P(B)}$
EQUATION Write the equation: Standard deviation (sample)	$s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n-1}}$
EQUATION Write the equation: Standard error of the mean	$\mathrm{SE} = \frac{s}{\sqrt{n}}$
KEY IDEA Complete the key idea: Exclusive events: add …	Exclusive events: add probabilities. Independent events: multiply

KEY IDEA Complete the key idea: Conditional probability: P(A|B) = …	Conditional probability: P(A|B) = P(A and B) / P(B)
KEY IDEA Complete the key idea: Normal distribution: 68% …	Normal distribution: 68% within 1σ, 95% within 2σ
KEY IDEA Complete the key idea: Standard error = σ/√n; …	Standard error = σ/√n; more data shrinks it slowly
QUICK CHECK For independent events A and B, P(A and B) is	P(A) × P(B)
QUICK CHECK About what percentage of a normal distribution lies within 1 SD of the mean?	68%
QUICK CHECK Taking 100 readings instead of 25 changes the standard error by a factor of	1/2
QUICK CHECK Physicists' '5-sigma' standard is	far stricter than p = 0.05
PROBLEM A fair coin is tossed three times. What is the probability of three heads? Give a decimal.	0.125. ½ × ½ × ½ = 1/8 = 0.125.
PROBLEM Two dice are rolled. What is the probability that the total is 7? Give a decimal.	0.1667. 6 ÷ 36 = 1/6 ≈ 0.167.
PROBLEM Find the mean of these timings in seconds: 2.1, 2.3, 1.9, 2.2, 2.0.	2.1 s. 10.5 ÷ 5 = 2.1 s.
PROBLEM A set of measurements has standard deviation 0.40 s. You take 16 readings. What is the standard error of the mean?	0.1 s. 0.40 ÷ √16 = 0.40 ÷ 4 = 0.10 s.

PROBLEM Heights are normally distributed with mean 170 cm and SD 10 cm. Roughly what fraction are between 150 cm and 190 cm?	95%. About 95%.
PROBLEM In a class, 60% study physics and, of those, 30% also study chemistry. What fraction of the class studies both? Give a decimal.	0.18. 0.6 × 0.3 = 0.18.
PROBLEM To halve the standard error of a mean, how many more readings do you need? Explain.	Four times as many. The standard error is s/√n, so it shrinks with the square root of n. To divide it by 2 you must multiply √n by 2, which means multiplying n by 4. Precision improves slowly with more data.

Module 1.6 · Vectors

EQUATION Write the equation: Magnitude	$\lvert\mathbf{a}\rvert = \sqrt{a_x^2 + a_y^2}$
EQUATION Write the equation: Direction	$\theta = \tan^{-1}\left(\frac{a_y}{a_x}\right)$
EQUATION Write the equation: Addition by components	$\mathbf{a} + \mathbf{b} = (a_x + b_x,\ \ a_y + b_y)$
EQUATION Write the equation: Dot product, components	$\mathbf{a} \cdot \mathbf{b} = a_x b_x + a_y b_y + a_z b_z$
EQUATION Write the equation: Dot product, geometric	$\mathbf{a} \cdot \mathbf{b} = \lvert\mathbf{a}\rvert\ \lvert\mathbf{b}\rvert \cos\theta$
KEY IDEA Complete the key idea: Vectors have magnitude …	Vectors have magnitude and direction; scalars only magnitude
KEY IDEA Complete the key idea: Magnitude = √(x² …	Magnitude = √(x² + y²); direction from $\tan^{-1}(y/x)$
KEY IDEA Complete the key idea: Add vectors …	Add vectors by adding components

KEY IDEA Complete the key idea: a · b = a_x b_x ...	a · b = a_x b_x + a_y b_y = |a||b| cos θ; zero when perpendicular
QUICK CHECK Displacement differs from distance because it	includes direction
QUICK CHECK (1, 2) + (3, −1) =	(4, 1)
QUICK CHECK The dot product of perpendicular vectors is	zero
QUICK CHECK A dot product is a	scalar
PROBLEM Find the magnitude of the vector (6, 8).	10. = 10.
PROBLEM Which of these is a vector?	velocity. Only velocity needs a direction.
PROBLEM A walker goes 3 km east then 4 km north. What is the size of her displacement?	5 km. √(9 + 16) = 5 km.
PROBLEM Add (2, −5) and (7, 1). What is the magnitude of the result?	9.849. √(81 + 16) = √97 ≈ 9.85.
PROBLEM A plane flies at 250 km/h on a heading 30° north of east. What is its northward velocity component?	125 km/h. 250 × sin 30° = 125 km/h.
PROBLEM Calculate the dot product of (3, −2, 4) and (1, 5, 2).	1. 3 − 10 + 8 = 1.
PROBLEM Two vectors have a dot product of zero, and neither is the zero vector. What can you conclude, and why?	They are perpendicular. The dot product equals |a||b|cos θ. Neither magnitude is zero, so cos θ must be zero, which means θ = 90°.

Module 1.7 · Mathematical thinking

EQUATION Write the equation: Sum of two even numbers	$2m + 2n = 2(m + n)$
EQUATION Write the equation: Root two, contradiction step	$\sqrt{2} = \frac{a}{b} \Rightarrow a^2 = 2b^2$
EQUATION Write the equation: Sum of first n integers (proved by pairing)	$1 + 2 + \cdots + n = \frac{n(n+1)}{2}$
KEY IDEA Complete the key idea: Theorems are proved from definitions …	Theorems are proved from definitions and axioms; they do not get overturned
KEY IDEA Complete the key idea: One counter-example disproves a …	One counter-example disproves a general claim; one example proves nothing
KEY IDEA Complete the key idea: Proof by contradiction: …	Proof by contradiction: assume the opposite, derive nonsense
KEY IDEA Complete the key idea: Every model has …	Every model has assumptions; know where they break
QUICK CHECK One example that agrees with a general claim	proves nothing in general
QUICK CHECK √2 is	irrational
QUICK CHECK Checking what a formula predicts when a quantity goes to zero is a	limiting case
QUICK CHECK Every model	leaves something out
PROBLEM Someone claims 'all prime numbers are odd'. What settles it?	The counter-example 2. 2 is prime and even.

PROBLEM Use the formula $n(n+1)/2$ to find $1 + 2 + \ldots + 100$.	5050. $100 \times 101 \div 2 = 5050$.
PROBLEM Prove that the product of two odd numbers is odd.	Write the odd numbers as $2m + 1$ and $2n + 1$. Their product is $4mn + 2m + 2n + 1 = 2(2mn + m + n) + 1$, which is one more than an even number, so it is odd.
PROBLEM A model of a falling stone ignores air resistance. For which object does this assumption fail most badly?	a feather. A feather is light with a large surface.
PROBLEM Exponential growth predicts a bacterial colony doubling forever. Which observation shows the model has broken down?	Growth slows as nutrients run out. Slowing growth signals that limit.
PROBLEM A student derives that the time for a ball to fall a height h is $t = \sqrt{(2h/g)}$. Check the formula in the limit $h \to 0$ and by units.	If $h = 0$ then $t = 0$, which is right: no fall takes no time. Units: h in m, g in m/s^2, so h/g has units s^2, and its square root is s, a time. Both checks pass.
PROBLEM In a proof by contradiction, you	assume the statement is false and derive an impossibility. That is the definition of the method.
PROBLEM Explain why a scientific theory can be overturned but a mathematical theorem cannot.	A theorem follows by logic from stated definitions and axioms, so it stays true as long as those assumptions hold; nothing observed in the world can change a deduction. A scientific theory makes claims about the world and is tested against evidence, so new observations can show it is wrong or limited, as Newton's mechanics was shown to fail at very high speeds.